INCH-POUND
MIL-DTL-24646/7C
w/AMENDMENT 1
17 June 2024
SUPERSEDING
MIL-DTL-24646/7C
29 January 2015

DETAIL SPECIFICATION SHEET

# FUSE BOXES, FEEDER DISTRIBUTION, SUBMERSIBLE (15 FEET), 125 VOLTS, ALTERNATING CURRENT, SINGLE PHASE BUSES, 0-30 AMPERE MIDGET FUSES, SYMBOLS 635.1, 653.1 AND 654.1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24646.

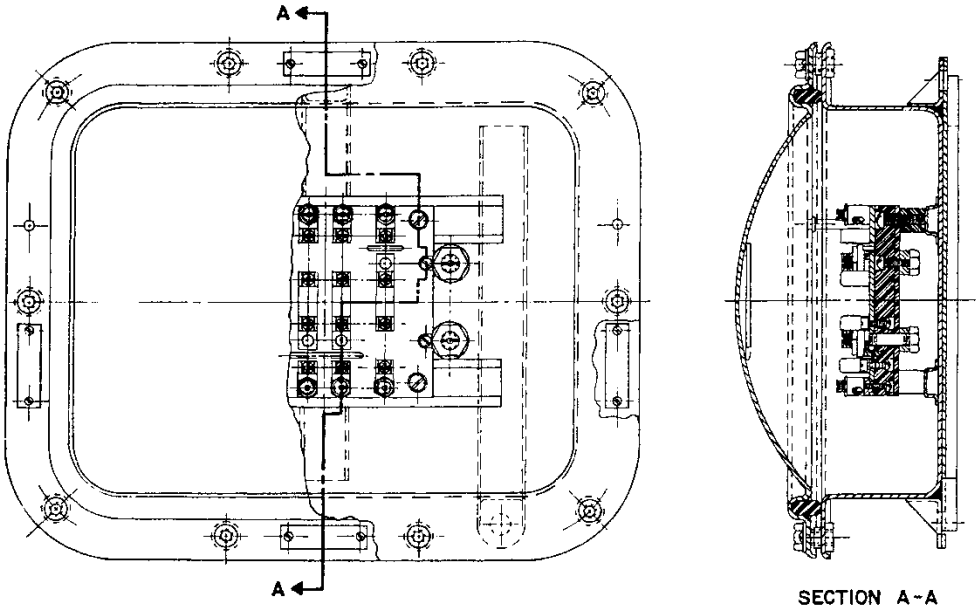


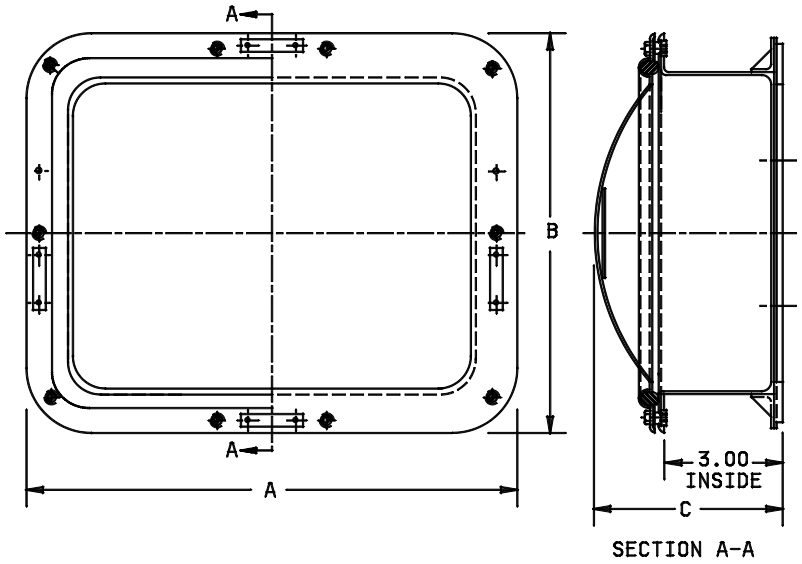

FIGURE 1. Fuse box assembly, typical. Item No. 01, 02, 03 (Symbol 635.1 shown)

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified herein, tolerances shall be as follows:
   Two place decimals - plus or minus 0.02.
   Three place decimals - plus or minus 0.005.

AMSC N/A
FSC 5920

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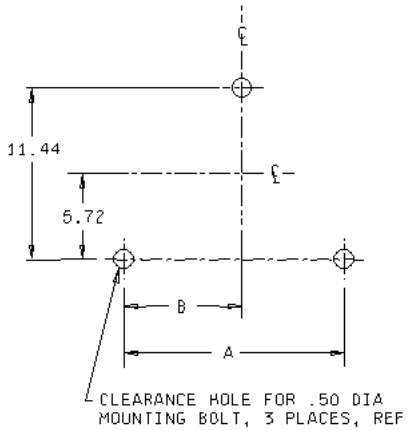



| Symbol No. | Dimension (ref ) | | |
|---|---|---|---|
| | A | B | C |
| 635.1 | 16.38 | 13.38 | 5.62 |
| 654.1 | 19.38 | 13.38 | 5.62 |
| 653.1 | 27.38 | 15.38 | 5.62 |

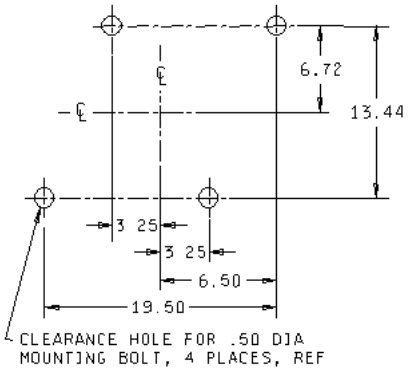
FIGURE 2. Enclosure, overall dimensions. Item No. 04, 05, 06
(Symbols 635.1, 654.1, and 653.1)



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A, Symbols 635.1, and 654.1.



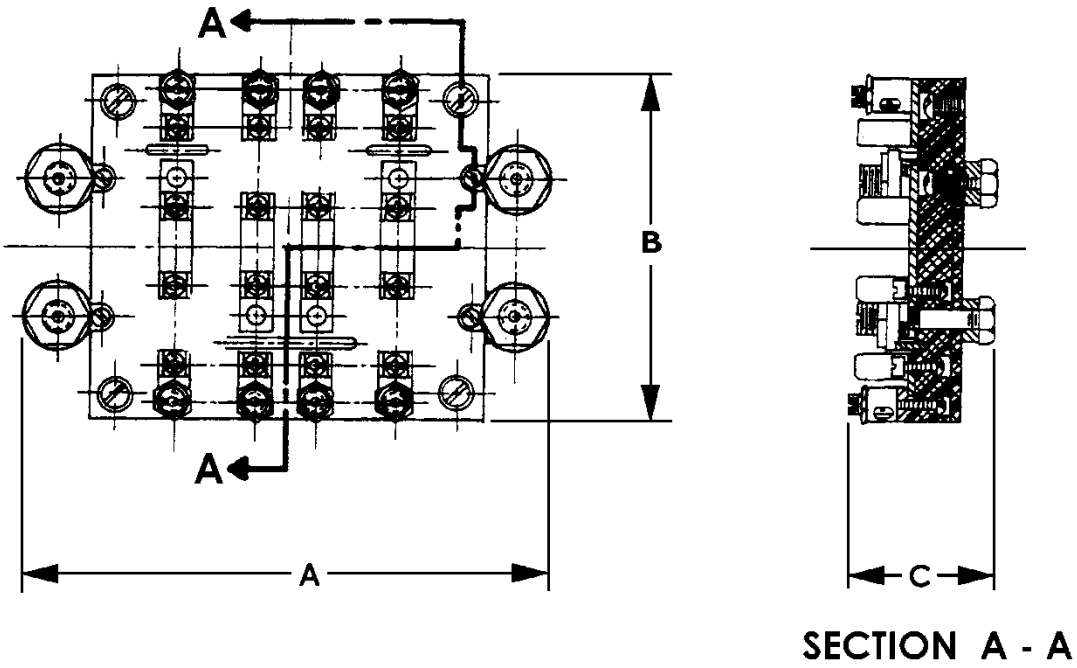

B. Symbol 653.1

| | DIMENSIONS | |
|---|---|---|
| | A | B |
| 635.1 | 9.25 | 4.62 |
| 654.1 | 11.75 | 5.88 |

FIGURE 3. Mounting dimensions, ref.



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SECTION A - A

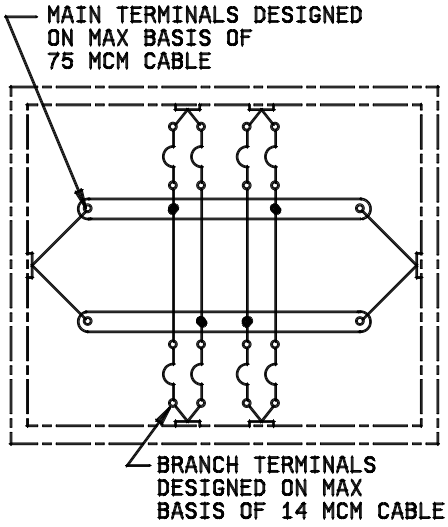
| SYMBOL NO | DIMENSIONS (MAX) | | |
|---|---|---|---|
| | A | B | C |
| 635.1 | 7.50 | 5.00 | 2.30 |
| 654.1 | 12.63 | 5.00 | 2.30 |
| 653.1 | 20.25 | 6.38 | 2.30 |

FIGURE 4. Fuse block assembly, typical. Item No. 07, 08, 09
(Symbols 635.1, 654.1, and 653.1)

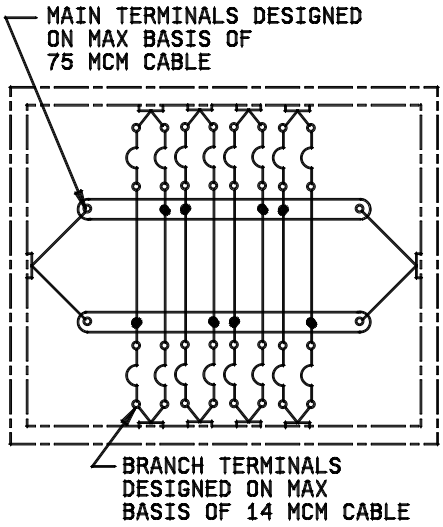


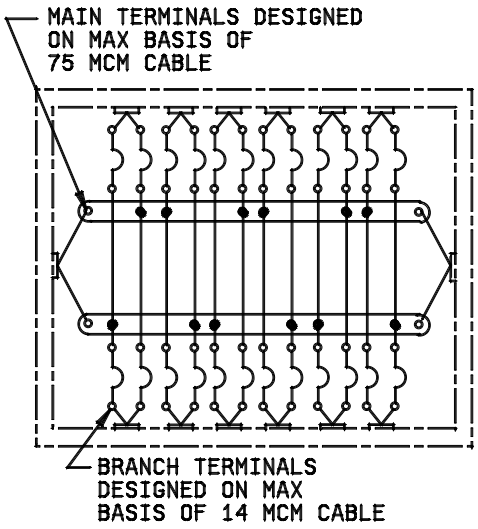
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A. Four single-phase branches
(Symbol 635.1)




B. Eight single-phase branches
(Symbol 654.1)




C. Twelve single-phase branches
(Symbol 653.1)

FIGURE 5. Wiring diagrams. Item No. 10, 11, 12



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## REQUIREMENTS

Symbol numbers: See table I.

Design:
Electrical rating: 125 volts alternating current, 0-30 ampere, 60 hertz.
Dimensions and configuration: See figures 1, 2, 3 and 4.
Wiring diagrams: See figure 5.
Mounting dimensions: See figure 3.
Degree of enclosure: Submersible (15 feet).
Designation and marking: See MIL-DTL-24646.
Parts: The parts listed in table II and specified in MIL-DTL-24646 shall be used in the construction of the fuse boxes specified herein.

General information:
Military part numbers and general information are as specified in table I.

TABLE I. Military part numbers and general information.

| Military part no. | Symbol no. | Approx. Weight in pounds 1/ | No. of circuits | Feeder MCM | Branch MCM | Fuse size | Bus bar size | Supersedes | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Drawing no. | Type |
| M24646/7 001 | 635.1 | 23.5 | 4 | 75 | 14 | midget 0-30A | 0.62 x 0.25 | 9-S-74200 | N/A |
| M24646/7-002 | 654.1 | 28.5 | 8 | 75 | 14 | midget 0-30A | 0.62 x 0.25 | 9-S-5305 | XXXY |
| M24646/7-003 | 653.1 | 52.6 | 12 | 75 | 14 | midget 0-30A | 0.62 x 0.25 | 9-S-5306 | XXXV |

1/ The weight of these fuse boxes may vary + or - 15 percent.



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TABLE II. Part list.

| Item no. 1/ | Description | QTY M /- 001 | 002 | 003 | Material/specification | Remarks |
|---|---|---|---|---|---|---|
| 01 | Fuse box assembly (sym 635.1) | - | | | MIL-DTL-24646/7 | See figure 1 |
| 02 | Fuse box assembly (sym 654.1) | | - | | MIL-DTL-24646/7 | See figure 1 |
| 03 | Fuse box assembly (sym 653.1) | | | - | MIL-DTL-24646/7 | See figure 1 |
| 2/ 04 | Enclosure (10 x 13 x 3) | 1 | | | M24142/6-004 | See figure 2 |
| 2/ 05 | Enclosure (10 x 16 x 3) | | 1 | | M24142/6-005 | See figure 2 |
| 2/ 06 | Enclosure (12 x 24 x 3) | | | 1 | M24142/6-010 | See figure 2 |
| 07 | Fuse block assembly | 1 | | | MIL-DTL-24646/7 | See figure 4 |
| 08 | Fuse block assembly | | 1 | | MIL-DTL-24646/7 | See figure 4 |
| 09 | Fuse block assembly | | | 1 | MIL-DTL-24646/7 | See figure 4 |
| 10 | Wiring diagram | 1 | | | MIL-DTL-2036 | See figure 5 |
| 11 | Wiring diagram | | 1 | | MIL-DTL-2036 | See figure 5 |
| 12 | Wiring diagram | | | 1 | MIL-DTL-2036 | See figure 5 |
| 13 | Fuse clip (0-30A) | 16 | 32 | 48 | A-A-55507/1-04 | N/A |
| 14 | Identification plate | 1 | 1 | 1 | Brass, ASTM B36/B36M B121/B121M | N/A |
| 15 | Information plate | 4 | 8 | 12 | Brass, ASTM B36/B36M B121/B121M | N/A |

1/ Item numbers by the figure titles are the same as the item numbers shown in this table.
2/ Enclosure is same as MIL-E-24142/6, except for inside depth which is 3 inches instead of 5 inches shown.



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Referenced documents. In addition to MIL-DTL-24646, this document references the following:

MIL-DTL-2036 MIL-E-24142/6 ASTM B36/B36M ASTM B121/B121M A-A-55507/1

The margins of this specification are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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